PHYSICS-INFORMED NEURAL NETWORKS · LORENZ INITIAL-VALUE PROBLEM

Sequential Causal Training of Windowed PINNs for the Lorenz ODE System

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27
equal time windows
over t ∈ [0, 13.26446]

6.97×10^{-5}
combined state RMSE
13,265 evaluation times

3.09×10^{-4}
maximum state-error norm
same evaluation grid

1.87×10^{-7}
ODE residual MSE
dense evaluation grid

FINDING In one saved, single-seed run, the windowed PINN followed a nearly closed Lorenz-system orbit with low state error against a DOP853 reference.

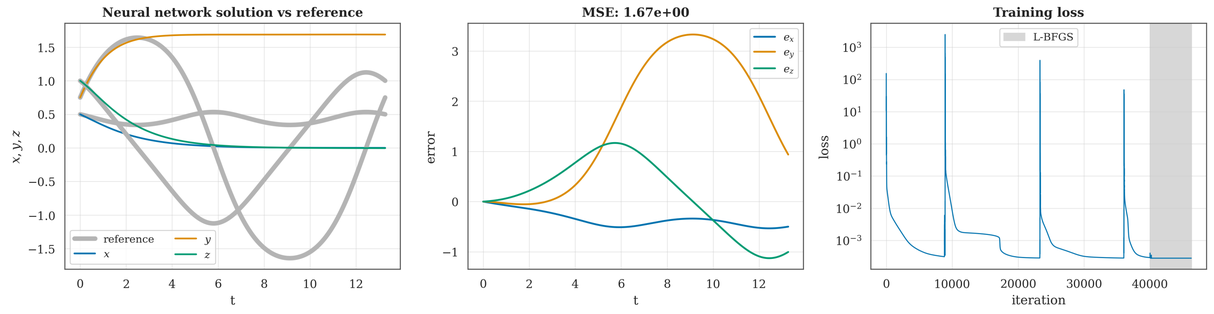
01 Problem setup

Approximate one nearly closed orbit of a three-state Lorenz initial-value problem.

$$\dot{x} = -0.1\,yz$$
$$\dot{y} = 1.6\,xz$$
$$\dot{z} = -0.75\,xy$$

$u(0) = (0.5, 0.75, 1.0)^T$
$t \in [0, 13.26446]$

02 Full-interval observation



The saved full-interval network settled away from the reference trajectory, despite a reduced training objective. Original paper figure.

03 Windowed PINN method

Initial-value problem
u(0) = (0.5, 0.75, 1.0)
t ∈ [0, 13.26446]

→

27 equal windows
≈ 0.491 time units
1,536 collocation times each

→

Window k PINN
Separate 4 × 60 tanh network
Hard initial condition

→

Train and refine
Four causal Adam stages
Unweighted L-BFGS

Repeat k = 0 … 26: save the trained window, pass its predicted endpoint as the next initial state, and warm-start the next network with its weights.

After training, compare the stitched prediction with DOP853. Reference values do not enter the optimization objective or stopping rule.

Inside each window

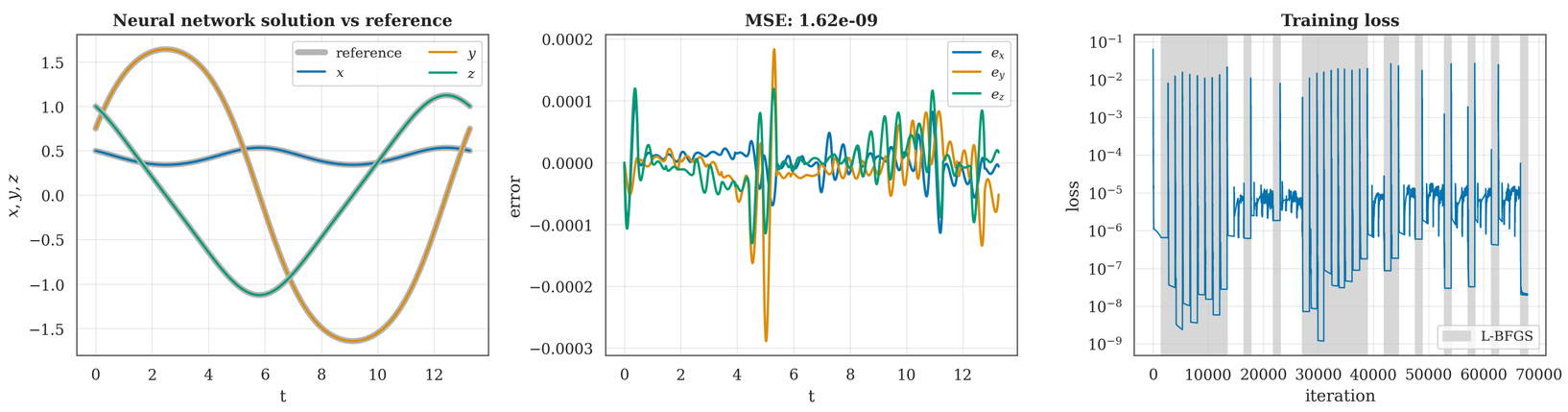
Hard initial condition
$\hat{u}_k(t) = u_{0,k} + (t - t_k)N_k(t)$

ODE residual
$r_k(t) = d\hat{u}_k/dt - f(\hat{u}_k(t))$

Causal Adam stages
ε = 0.01 → 0.1 → 1 → 10; advance when min weight > 0.99 or at 4,000 updates.

Refine and hand off
Unweighted L-BFGS, then pass the predicted endpoint state and warm-start weights.

04 Trajectory and error evidence



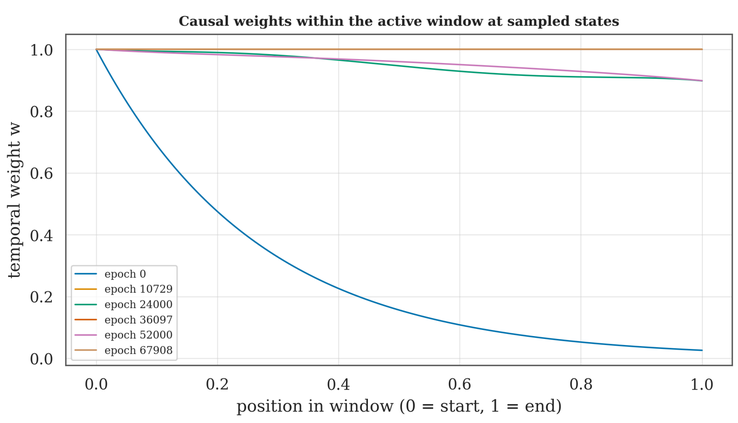
Saved 1536-point candidate run: prediction against the numerical reference, signed state errors, and training-loss history. Headline state-error metrics use the 13,265-time evaluation grid.

05 Measured results

Measure	Full interval	27 windows
Networks	1	27
Collocation points	39,793 LHS	41,446 uniform
Combined state RMSE	2.237	6.97×10^{-5}
Max state-error norm	3.346	3.09×10^{-4}
Residual MSE collocation grid	2.80×10^{-4}	1.87×10^{-7}
Recorded wall time	4,529 s	881 s

Descriptive comparison of two saved configurations. Network count, sampling, loss, and optimizer schedule also differ, so this does not isolate the effect of causal weighting.

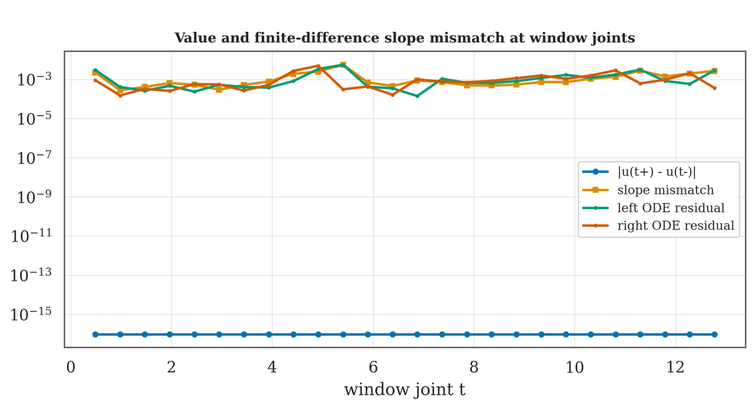
Causal weights



Time-ordered residual weights emphasize earlier points before later points in each window.

Candidate 1536-point run · ε stages 0.01, 0.1, 1, 10.

Window handoff



At each interface, the next trial solution starts from the previous predicted endpoint. State values are continuous by construction.

Derivative continuity is not enforced.

06 Scope and limitations

One saved seed, initial state, coefficient set, and nearly closed interval.

Earlier prediction error can propagate through endpoint handoff.

State values meet at interfaces; derivatives may jump.

No saved matched windowed run without causal weights.

The saved run demonstrates this implementation on this Lorenz initial-value problem. A matched, multi-seed study is needed to isolate component effects and test broader reliability.